MSGR-RAG: Reviewer-Response Status and Remaining Plan

Reviewer 2 and Reviewer 3 comments on the revised manuscript (R2). Deadline for resubmission: 14 November. All numbers below were computed from the frozen result files in `msgr-rag-v1/Datasets/intermediate/`; scripts and outputs are in `msgr-rag-v1/reviewer_response/`. The headline confusion matrix (TP 574, FP 85, FN 0, TN 315; accuracy 91.27%, F1 93.11%) was re-asserted before any new number was produced and did not change.

1. Summary

- **The core result stands.** On held-out halves of the 974 facts (thresholds re-chosen on the other half, 20 random splits) the full rule scores 91.2% accuracy (sd 1.1) and 93.0% F1. The reported 91.3% is not an artefact of threshold selection.
- **The reviewers are right about the test set.** A plain entity-level lookup (S_E < 1) already scores 100% on all 974 facts, and the domain-inconsistent labels follow the same K=2 rule the domain scale applies. The paper must say this and shrink its claims to this evaluation setting.
- **The full rule is a deliberately conservative, safety-oriented operating point.** It misses no hallucination on any of the 974 facts (0 false negatives), whereas a confidence-only rule at the pre-set threshold 0.55 misses 3 and reaches 100% only at a threshold chosen after seeing the labels. The price is 85 true facts routed to review (precision 87.1%), and every one is traceable to a named score. The ablation (Section 4.1) is reported in full because the reviewers asked for it.
- **There is independent evidence for a multi-scale benefit** (edge-masking test, Section 4): when entity edges are missing, concept and domain scores recover true facts (AUC about 0.88 combined) where the entity score cannot. The shipped decision rule does not yet use that signal; this is the main future-work direction.
- **Against plain RAG, MSGR-RAG is better in specific areas, not overall.** Predicate-swapped questions: 39.0% vs 25.3%, significant (p = 0.0025). Domain-inconsistent: 100% vs 94.0%, directionally better but not significant (n = 50). Overall and on real relationships plain RAG is higher. Section 3 gives the framing and the exact paper edits.
- **Two items must be fixed before resubmission:** (a) Table 16 / Fig. 7 plain-RAG numbers, whose provenance has to be documented or replaced (Section 5); (b) the gate attribution in Section 5.6 of the paper.

2. Response status by reviewer comment

Comment	What we did / found	Change in the paper	Status
R2-W1, Q1, S1 / R3 major 1: labels depend on K=2	Confirmed: entity-only = 100%; all 75 domain-inconsistent facts have S_E = 0.1 by construction. Ran edge-masking test (hide 10-30% of real entity edges, rebuild concept/domain graphs; no K-derived label): S_C+S_D AUC 0.86-0.88 vs S_E 0.33.	Sec 4.3, 5.4, 8, abstract: state that the 100% checks the set's construction, not unseen-case detection. Add edge-masking table and paragraph to Sec 5.	Experiment done; text pending
R2-W2, Q3, S3: thresholds chosen on the same data	Thresholds came from the sweep on the 981-fact pre-refinement set. Nested 50/50 split x 20: acc 91.2 +/- 1.1, prec 87.0 +/- 1.4, rec 100, F1 93.0 +/- 0.8. Chosen thresholds (0.50, 0.55) in 20/20 splits.	Sec 5.1/5.4: say where thresholds were chosen; add nested-split row to Tables 9-10. Keep 91.3% as full-set figure.	Done; text pending

Comment	What we did / found	Change in the paper	Status
R2-W3, Q2, S2 / R3 major 2: no ablation, confidence-only	Ablation on identical facts: entity-only 100.0%; confidence-only 99.69% (0 FP, 3 FN); full rule 91.27%; consistency gate removed: identical; pattern gate removed: 78 FP. AUC: S_E 1.000, S_C 0.887, S_D 0.715, C(f) 0.807, Conf(f) 1.000.	Add ablation table (Sec 5.3/5.7). Rewrite 5.7 around the operating-point trade-off: the full rule has zero missed hallucinations (confidence-only at the pre-set 0.55 misses 3), at a measured, traceable cost of 85 flagged true facts; say that a confidence-only rule is a higher-precision alternative when a missed error is less costly. Fix 5.6: pattern I3 alone flags all 85 FPs (78 also fall under the consistency threshold); delete the 'non-overlapping subsets' claim.	Done; text pending
R2-W4, S5 / R3: 'Hallucination-Resilient' not supported; practical readiness	Pipeline: 91.3% (clean triples), 75.7% on the 75.3% resolved, 57.0% over all 974 (UNKNOWN counted as error).	Drop 'Hallucination-Resilient' from the title; report all three numbers in the abstract; remove 'robust'/'ready' wording in 6.5 and 8.	Text pending
R2-W5, Q4, S4: counts do not reconcile	202 candidates = 90 admitted + 112 withheld (text correct). Domain graph 119 edges = 90 K=2-abstracted + 29 hand-curated. 149 entities = 95 drugs + 42 conditions + 12 adverse effects (5 cohort drugs yield no relationship). 103,552 relationships collapse to 103,551 DiGraph edges. 187 to 118 FPs = 36.9% reduction.	Sec 3.3 and Table 4: add '119 = 90 + 29 expert-added'. Sec 4.2: explain 95 drugs and the one collapsed edge. Sec 5.2: replace 'nearly halves' by '36.9% reduction'. Fix Eq. 14/21 cross-reference; define the satisfiability symbol, divergence, P().	Verified; text pending
R2-W6, S5: refs [14], [20], [35]	Text search of both PDFs. [14] (arXiv 2311.07914): none of 'multi-scale', 'level of abstraction', 'single level', 'granularity' occurs. [20] (arXiv 2408.08921): 'granularity' = retrieved unit (nodes, triplets, paths, subgraphs); its future-prospects section lists no multi-scale verification. [35] is BM25.	Sec 2.2, 1.1, abstract: remove the attributions; rewrite as the authors' own observation ('to our knowledge, the surveyed KG-grounded methods do not perform cross-granularity consistency checks'). Call [35] BM25.	Verified; text pending
R2-W7, S5: weak baseline	Added closed-book gemma3 judge on the same 974 triples: acc 61.7%, F1 69.4% vs MSGR 91.3% / 93.1% (McNemar p = 1e-49). TF-IDF (914 aligned facts): 68.6%.	Sec 4.4/5.5: add judge row; state it is closed-book and one 4B-class model; list NLI, RAGAS, ARES, RAGTruth under future work.	One baseline done; others future
R3: NL degradation, tiny regex test, inconsistent denominators	Re-computed plain RAG and MSGR on the same 733 resolved questions and on all 974 (Section 3).	Replace Table 16 with the aligned table; label Sec 6.7 'anecdotal, n = 5'; attribute the predicate-swap drop to question generation and list it as a pipeline limitation.	Done; text pending
R3 minor: abstract too long; deterministic vs learned verifier; generalisation	No experiment needed.	Cut abstract to about 150 words. Add a scope paragraph to Sec 1.3. Label Eq. (5) 'not evaluated'. Sec 8: one DrugBank cohort, one LLM.	Text pending

3. MSGR-RAG versus plain RAG: where it excels

Plain RAG is the primary measured run (`simple_rag_v2_results.csv`): the same gemma3 model, the same 974 questions, the entity's first five known facts as context, and the model's own VERIFIED/HALLUCINATION label. Both systems are scored on the 733 questions the MSGR pipeline resolves. Significance is an exact McNemar (sign) test on the discordant pairs.

Category (resolved subset)	n	MSGR-RAG	Plain RAG	MSGR right / plain wrong	Plain right / MSGR wrong	p-value	Reading
Predicate-swapped	154	39.0%	25.3%	33	12	0.0025	MSGR better, significant
Domain-inconsistent	50	100.0%	94.0%	3	0	0.25	MSGR higher; not significant (small n)
Absent relationship	170	98.2%	100.0%	0	3	0.25	Tie
Real relationship	359	77.4%	89.7%	23	67	3.8e-6	Plain RAG better, significant
Overall	733	75.7%	78.9%	59	82	0.064	Statistically tied

What can honestly be emphasised

- **Predicate-swapped facts (the patient-safety error, e.g. 'causes' for 'treats').** MSGR-RAG is significantly better than plain RAG (39.0% vs 25.3%, p = 0.0025), and on clean triples the verification core catches 100% of them against 83.0% for the closed-book LLM judge and 0.0% for TF-IDF. The absolute pipeline figure is low because the LLM question generator drifts to an interaction phrasing (91 of 94 resolved errors in Section 6.4 of the paper); that limitation applies to both systems.
- **Domain-inconsistent facts.** 100% vs 94.0% in the same direction, but 3 versus 0 discordant pairs on 50 questions is not significant. Report it as 'higher, not significant', not as a win.
- **Abstention with a traceable reason.** MSGR-RAG returns UNKNOWN for 24.4% of questions (entity outside graph coverage) and each decision cites graph edges and scores. Plain RAG has no such mechanism. Plain RAG scores 82.6% on those 241 questions MSGR abstains on, so coverage, not correctness, is the price of abstaining.
- **Auditability.** Every verdict decomposes into S_E, S_C, S_D, C(f) and Conf(f); every false positive can be traced to a gate (Sec 5.6 of the paper). This is a structural property of the method, not a benchmark number.
- **Honest weaknesses to state alongside:** lower overall accuracy than plain RAG, significantly lower on real relationships (77.4% vs 89.7%), and 57.0% over all 974 questions when UNKNOWN counts as an error (plain RAG: 79.8%). Plain RAG's retrieval here is the first five facts of the subject, not question-relevant ranking, so it is a weak but fair-by-construction baseline; say so.

Exact edits to the paper for this comparison

Where	Edit
Abstract	Replace 'outperforms a plain retrieval-and-generation baseline on every category, most substantially on domain-inconsistent facts (+19.5 points)' with: 'Against a plain RAG baseline on the 733 resolved questions, MSGR-RAG is significantly more accurate on predicate-swapped questions (39.0% vs 25.3%, p = 0.0025), higher but not significantly so on domain-inconsistent questions (100% vs 94.0%), and less accurate on real relationships (77.4% vs 89.7%); overall the two are statistically tied (75.7% vs 78.9%, p = 0.064).'
Sec 1.3 (last contribution)	Say 'compare against a plain retrieval-and-generation baseline, identifying the categories where verification helps and where it costs accuracy' instead of 'diagnosing the one substantial gap'.
Sec 6.6 and Table 16	Replace the table with the aligned table above (and all-974 figures: MSGR 57.0% with UNKNOWN as error, plain RAG 79.8%). Delete 'improves accuracy ... on every category'. Fig. 7: redraw from the same numbers.
Sec 6.5	Delete 'robust'. Say: 'on resolved questions the core's category pattern is preserved for three of four categories; the exception is predicate-swapped (39.0%), caused by question generation.'
Sec 8 (Conclusion)	Replace 'improves over a plain RAG baseline on every category' with the abstract sentence above; add one sentence on abstention (24.4% UNKNOWN) and traceability.

Where	Edit
Sec 7 (limitations)	Add: single LLM (gemma3), baseline retrieval is the first five entity facts, one DrugBank cohort, category tests on small n (domain-inconsistent n = 50).

4. Evidence added in this revision

4.1 Ablation (identical 974 facts, thresholds 0.50 / 0.55)

Rule	FP	FN	Accuracy	Precision	Recall	F1
Entity-only (S_E < 1)	0	0	100.0%	100.0%	100.0%	100.0%
Confidence-only (0.55)	0	3	99.69%	100.0%	99.48%	99.74%
Confidence-only (gap midpoint 0.687, oracle)	0	0	100.0%	100.0%	100.0%	100.0%
Full rule (C | pattern | Conf) - paper	85	0	91.27%	87.10%	100.0%	93.11%
Full minus consistency gate	85	0	91.27%	87.10%	100.0%	93.11%
Full minus pattern gate	78	0	91.99%	88.04%	100.0%	93.64%
Concept-only (S_C < 1)	7	123	86.65%	98.47%	78.57%	87.40%
Domain-only (S_D < 1)	111	145	73.72%	79.44%	74.74%	77.02%

4.2 Edge masking (independent of the K=2 rule)

Hide a random fraction of the 400 real entity edges, rebuild the concept and domain graphs from the masked entity graph with the same K=2 rule plus the 29 hand-curated edges, then ask whether the scores separate the held-out TRUE facts from the 574 false facts among facts the entity scale cannot support. 5 seeds per rate. The scorer is a re-implementation validated against the frozen scores (S_E 100%, S_C 99.9%, S_D 97.5% agreement); masked and unmasked runs use the same scorer.

Held out	AUC S_E	AUC S_C	AUC S_D	AUC S_C+S_D	AUC Conf	AUC C(f)	Held-out true flagged by full rule
10%	0.328	0.828	0.674	0.856	0.752	0.116	100%
20%	0.330	0.835	0.704	0.876	0.772	0.101	100%
30%	0.331	0.841	0.698	0.880	0.770	0.098	100%

Reading: concept and domain scores carry real signal where the entity scale is blind, which is independent support for a multi-scale contribution. The shipped rule discards it: C(f) is anti-correlated with truth in this setting (AUC about 0.10) and the full rule flags every held-out true fact. A crude accept rule (S_C = 1 and S_D = 1) recovers 62-63% of held-out true facts at a 5.2% false-accept rate; present this only as exploratory and as the main future-work direction.

4.3 Baselines on the same 974 clean triples

System	n	Accuracy	Precision	Recall	F1
MSGR-RAG core (full rule)	974	91.3%	87.1%	100.0%	93.1%
Closed-book gemma3 judge (temperature 0)	974	61.7%	65.6%	73.5%	69.4%
TF-IDF similarity (pre-refinement set, aligned)	914	68.6%	100.0%	44.2%	61.3%

The judge is closed-book (no knowledge-graph evidence) and a single small model: it shows the task is not solved by a general LLM, not that MSGR-RAG beats a retrieval-grounded judge.

5. Items that must be resolved before resubmission

Item	Why it matters	Action
Provenance of the Table 16 plain-RAG column	The paper's plain-RAG numbers (67.8%; 84.4 / 71.6 / 80.5 / 30.6) equal the column `simple_rag_decision` in `plain_rag_sensitivity.csv`. That file differs from the primary run (`simple_rag_v2_results.csv`) in 117 decisions with identical answer text, and no code, notebook or commit in the repository produces it. A reviewer who opens the data will ask how those decisions were produced.	Author to document how the 117 decisions were generated (script or procedure, placed in the repository). Report the primary run in Table 16 as the main comparison; if the sensitivity variant is kept, report it separately with its method described. Plan: rerun cells 8-9 of `semantic_rag_with_tf_idf.ipynb` to a new output file and compare row by row.
Gate attribution (Sec 5.6)	Pattern I3 alone flags all 85 FPs; the '78 + 7 non-overlapping' split depends on check order.	Rewrite as in Section 2.
Reverse-direction rule	The paper says interacts_with checks the reverse direction; my re-implementation matches the frozen scores only if the concept scale is forward-only, and 24 domain-scale cases remain unexplained.	Check the original Component 5 code path for the exact rule; make the paper's Table 3 caption match it.
Stale decision column	`verification_results.csv` holds decisions at thresholds 0.80 / 0.50; the paper reports 0.50 / 0.55.	Regenerate the column or note it in the repository README.
Table 3 'other edge 0.6'	This value never occurs in the 974-fact data.	Remove it or state it is unused.
Domain-inconsistent vs S_D	15 of 75 such facts have S_D = 1.0 under the ATC-grounded mapping (category defined under the transitive mapping).	State the definition precisely in Sec 4.3.

6. Future scope (accepted as outside this revision)

- **Use the concept/domain signal in the decision rule.** The edge-masking result shows a rule that accepts when concept and domain both corroborate can recover true facts the entity scale misses. A principled version (learned or calibrated combination, validated on held-out edges) is the main way to demonstrate a multi-scale gain.
- **Learned verifier (Eq. 5)** with negative sampling independent of the evaluation-set construction rules.
- **Independent labels** for domain-inconsistent facts (for example expert review or a clinical ontology other than the K=2 rule).
- **Stronger baselines:** NLI-based faithfulness models, RAGAS, ARES, RAGTruth-trained detectors, and a retrieval-grounded LLM judge; plain RAG with question-relevant retrieval.
- **Generalisation:** a second LLM for extraction and plain RAG, a second domain or dataset (for example SIDER, BioASQ, PubMedQA), and multi-hop questions.
- **Extraction from genuine text:** replace the regex extractor with a parser- or model-based extractor and evaluate on more than five sentences.
- **Sensitivity studies** for the Table 3 floors, the confidence weights, and K.

7. Suggested order of work to 14 November

Step	Task	Effort
1	Document or replace the plain-RAG sensitivity column; replace Table 16 and Fig. 7 (Section 3).	half a day
2	Rewrite Sec 5.6, 5.7, 4.3, 5.4 and the abstract/conclusion (Sections 2-3).	1 day
3	Mechanical fixes: title, Sec 3.3 / Table 4 counts, Eq. cross-reference, symbol definitions, refs [14] [20] [35], abstract length.	2-3 hours
4	Add three tables to Sec 5: nested split, ablation, edge masking; add judge row to Sec 5.5.	half a day
5	Write the response letter point by point from Section 2 (concede circularity and the confidence-only result; cite the new tables).	half a day
6	Optional: rerun plain RAG into a new file and compare; check the original Component 5 reverse-direction rule.	1-2 hours each

Appendix: reproduction

From `msgr-rag-v1/reviewer_response/`: `python3 -I analyses.py` (ablation, nested split, aligned denominators, count reconciliation); `python3 -I edge_masking.py`; `python3 -I judge_eval.py` (judge predictions already saved; `llm_judge.py` regenerates them in about 10 minutes). Outputs are in `out/`. Narrative notes: `FINDINGS.md`, `MODIFICATION_REPORT.md`. The per-category plain-RAG significance test in Section 3 was computed directly from `component6_e2e_results.csv` and `simple_rag_v2_results.csv`.